Complete Beginner’s Guide: Testing UDP on ESP32 (Windows)

This step-by-step guide is designed for complete beginners with zero prior experience using an ESP32. By the end of this tutorial, your ESP32 will act as a UDP Echo Server, receiving data sent over Wi-Fi from your Windows PC and replying back instantly.

Prerequisites & Checklist

Hardware Needed

1. **ESP32 Development Board** (e.g., ESP32 Dev Module / NodeMCU-32S).
2. **Micro-USB or USB-C Data Cable** (Ensure it supports data transfer, not just power charging).
3. **Wi-Fi Network** (Both your PC and ESP32 must be connected to the same local network).

Software Needed

1. **Arduino IDE** (Used to upload code to the ESP32).
2. **Packet Sender** (Used on Windows to send and receive UDP packets).

Step 1: Install Packet Sender via Windows Terminal

1. Press Win + X on your keyboard and select **Terminal** or **PowerShell**.
2. Type the following command and press **Enter**:

DOS
winget install packetsender

3. If prompted to agree to source terms, type Y and press **Enter**. Windows will automatically download and install Packet Sender.
4. Once completed, search for **Packet Sender** in your Windows Start Menu to confirm it installed.

Step 2: Set Up Arduino IDE for ESP32

If you haven't set up your Arduino IDE for ESP32 yet:

1. Download and install **Arduino IDE 2.x** from arduino.cc.
2. Open Arduino IDE, go to **File > Preferences**.
3. In the field labeled **Additional Boards Manager URLs**, paste this URL: https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

4. Click **OK**.
5. Go to **Tools > Board > Boards Manager...**, type esp32 in the search bar, find **esp32 by Espressif Systems**, and click **Install**.

Step 3: Write and Upload the UDP Echo Code

1. Connect your ESP32 board to your computer using your USB cable.
2. In Arduino IDE, erase any existing code in the editor and paste the following program:

   [Use the file of UDP.ino]

3. Replace "YOUR_WIFI_SSID" and "YOUR_WIFI_PASSWORD" with your Wi-Fi name and password.
4. Select your board: Go to **Tools > Board > ESP32 Arduino > ESP32 Dev Module**.
5. Select your port: Go to **Tools > Port** and pick your ESP32's COM port (e.g., COM3 or COM4).
6. Click the **Upload** arrow icon (top-left). Wait until the bottom panel displays Done uploading.

Step 4: Obtain the ESP32 IP Address

1. Open the Serial Monitor in Arduino IDE (**Tools > Serial Monitor** or press Ctrl + Shift + M).
2. Set the baud rate dropdown in the bottom-right corner of the Serial Monitor to **115200 baud**.
3. Press the physical **EN** or **RST** button on your ESP32 board to restart it.
4. Watch the Serial Monitor output. You will see output similar to this:

Connecting to Wi-Fi....
WiFi Connected!
ESP32 IP Address: 192.168.1.50
Listening for UDP packets on port 4210

5. **Write down the IP address** (e.g., 192.168.1.50).

Step 5: Test UDP Communication using Packet Sender

1. Launch **Packet Sender** from your Start Menu.
2. At the top of the interface, configure the following fields:

Field	Value	Description
ASCII	Hello ESP32	The message you want to send

Address	192.168.1.50	Your ESP32's IP Address (from Step 4)
Port	4210	The UDP port configured in the code
Protocol	UDP	Select from the rightmost drop-down menu

3. Click the blue **Send** button on the right side.
4. *Firewall Notice:* If Windows Defender Firewall displays a prompt, check **Private networks** and click **Allow access**.

Step 6: Verify Success

In Packet Sender (Windows PC)

Look at the **Traffic Log** table at the bottom of Packet Sender:

- You will see an **Outgoing** row showing your sent message Hello ESP32.
- Within a few milliseconds, an **Incoming** row will appear with the message: ESP32 Echo: Hello ESP32

In Arduino Serial Monitor (ESP32)

Your Serial Monitor will log the transaction in real time:

Plaintext
Received 11 bytes from 192.168.1.15:52410
Payload: Hello ESP32

Troubleshooting Guide

- **No response in Packet Sender?**
  - Confirm that your PC and ESP32 are connected to the exact same Wi-Fi router (some dual-band routers isolate 2.4 GHz and 5 GHz networks).
  - Double-check that the IP address entered in Packet Sender matches the one shown in the Serial Monitor.
- **Serial Monitor shows gibberish characters?**
  - Ensure the speed in the bottom-right corner of the Serial Monitor is set to **115200 baud**.
- **Failed to upload / A fatal error occurred: Failed to connect to ESP32?**
  - Press and hold the **BOOT** button on your ESP32 board when the Arduino IDE displays Connecting..., then release it once uploading starts.